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SOLAR CELL AND PHOTOVOLTAIC MODULE

Abstract

The solar cell includes: a substrate; a tunneling dielectric layer disposed over a first surface of the substrate; a plurality of doped conductive layers arranged at intervals over the tunneling dielectric layer; a plurality of first electrodes each extending in a first direction, where the plurality of first electrodes are arranged at intervals along a second direction, and each first electrode is disposed on and electrically connected to a corresponding one of the plurality of doped conductive layers; and at least one conductive transport layer, where the at least one conductive transport layer includes a respective conductive transport layer between every two adjacent doped conductive layers and in contact with a side surface of each of the two adjacent doped conductive layers. The at least one conductive transport layer and the plurality of doped conductive layers are doped with doping ions of a same type.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation of U.S. patent application Ser. No. 18/600,467, filed on Mar. 8, 2024, which is a continuation of U.S. patent application Ser. No. 17/960,678, filed on Oct. 5, 2022, which claims the benefit of priority under the Paris Convention to Chinese Patent Application No. 202210940296.X filed on Aug. 5, 2022 and Chinese Patent Application No. 202222078085.8 filed on Aug. 5, 2022, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] Embodiments of the present disclosure relate to the field of solar cells, and in particular to a solar cell and a photovoltaic module.

BACKGROUND

[0003] A solar cell has desirable photoelectric conversion capability. Generally, a tunneling dielectric layer and a doped conductive layer are prepared on a surface of a substrate to suppress the carrier recombination on the surface of the substrate in the solar cell and enhance the passivation effect on the substrate. The tunneling dielectric layer has better chemical passivation effect, and the doped conductive layer has better field passivation effect. In addition, in order to transport and collect photogenerated carriers generated by the solar cell, electrodes are also prepared on a part of the surface of the substrate.

[0004] However, currently, the solar cell has the problem of low photoelectric conversion efficiency.

SUMMARY

[0005] A solar cell is provided according to an embodiment of the present disclosure. The solar cell includes a substrate; a tunneling dielectric layer disposed over a first surface of the substrate; a plurality of doped conductive layers arranged at intervals over the tunneling dielectric layer; a plurality of first electrodes each extending in a first direction, where the plurality of first electrodes are arranged at intervals along a second direction perpendicular to the first direction, and each first electrode of the plurality of first electrodes is disposed on and electrically connected to a corresponding one of the plurality of doped conductive layers; and at least one conductive transport layer, where the at least one conductive transport layer includes a respective conductive transport layer between every two adjacent doped conductive layers in the plurality of doped conductive layers, and the respective conductive transport layer is in contact with a side surface of each of the two adjacent doped conductive layers. The at least one conductive transport layer and the plurality of doped conductive layers are doped with doping ions of a same type.

[0006] In some embodiments, the respective conductive transport layer is disposed between a

corresponding pair of adjacent first electrodes of the plurality of first electrodes, and is spaced apart from each of the corresponding pair of adjacent first electrodes.

[0007] In some embodiments, each first electrode is partially embedded in the corresponding one of the plurality of doped conductive layers.

[0008] In some embodiments, there are a plurality of conductive transport layers disposed at intervals along the first direction.

[0009] In some embodiments, the at least one conductive transport layer is configured as a plurality of conductive transport layers, the plurality of conductive transport layers are disposed in a form of an array, the array comprises a plurality of columns of conductive transport layers disposed at intervals along the second direction, conductive transport layers in each column of the plurality of columns of conductive transport layers are disposed at intervals along the first direction, and there is at least one first electrode between two adjacent columns of conductive transport layers along the second direction.

[0010] In some embodiments, conductive transport layers in adjacent columns of conductive transport layers are arranged in a staggered manner along the first direction.

[0011] In some embodiments, each of a column of conductive transport layers is in one-to-one correspondence to each of an adjacent column of conductive transport layers, and every two adjacent conductive transport layers corresponding to each other are disposed in the spaced manner along the second direction.

[0012] In some embodiments, the solar cell further includes: a plurality of second electrodes disposed at intervals, where each of the plurality of second electrodes extends along the second direction and is electrically connected to the plurality of first electrodes disposed at intervals along the second direction.

[0013] In some embodiments, in a column of the conductive transport layers, there is at least one second electrode between two adjacent conductive transport layers.

[0014] In some embodiments, in the column of the conductive transport layers, there are two second electrodes between two adjacent conductive transport layers.

[0015] In some embodiments, conductive transport layers in adjacent columns of conductive transport layers are arranged in a staggered manner along the first direction, and two conductive transport layers belong to different columns of conductive transport layers and disposed in a stagger manner are located on opposite sides of the second electrode, respectively.

[0016] In some embodiments, the substrate comprises a peripheral area and a central area, the peripheral area is defined as a periphery of the second electrode located at an outermost side, the central area is defined as an area of the substrate apart from the peripheral area, and a distance between every two adjacent conductive transport layers located in the peripheral area in the first direction is smaller than a distance between every two adjacent conductive transport layers located in the central area in the first direction.

[0017] In some embodiments, among conductive transport layers in each column of conductive transport layers, there are multiple conductive transport layers in the peripheral area, and there is at most one conductive transport layer between two adjacent second electrodes in the central area.

[0018] In some embodiments, in each column of the conductive transport layers located in the central area, a distance between every two adjacent conductive transport layers in the first direction is constant.

[0019] In some embodiments, in each column of the conductive transport layers located in the central area, the distance between every two adjacent conductive transport layers ranges from 0.01 mm to 20 mm, in each column of the conductive transport layers located in the peripheral area, a distance between every two adjacent conductive transport layers ranges from 0.005 mm to 18 mm.

[0020] In some embodiments, the solar cell further includes a first connecting portion, where the first connecting portion is located between two adjacent conductive transport layers disposed at intervals along the first direction, and is electrically connected to side surfaces of the two adjacent

conductive transport layers.

[0021] In some embodiments, in the direction perpendicular to the surface of the substrate, a height of a respective one of the at least one conductive transport layer is 0.5 to 1.2 times a height of a corresponding one of the plurality of doped conductive layers.

[0022] In some embodiments, a top surface of the at least one conductive transport layer has a light trapping structure, and the light trapping structure includes multiple pyramid structures, each of the multiple pyramid structures has a bottom surface and a side surface connected to the bottom surface.

[0023] In some embodiments, the solar cell further includes a first passivation layer, one part of the first passivation layer covers the first surface of the substrate, and the other part of the first passivation layer covers top surfaces of the plurality of doped conductive layers and a top surface of the at least one conductive transport layer.

[0024] In some embodiments, a photovoltaic module is provided and includes at least one cell string, where each of the at least one cell string is formed by connecting a plurality of solar cells, each of the plurality of solar cells being a solar cell described above; at least one package layer configured to cover a surface of the at least one cell string; and at least one cover plate configured to cover a surface of the at least one package layer away from the at least one cell string.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] One or more embodiments are described as examples with reference to the corresponding figures in the accompanying drawings, and the exemplary description does not constitute a limitation to the embodiments. The figures in the accompanying drawings do not constitute a proportion limitation unless otherwise stated.

[0026] FIG. 1 is a schematic structural view of a top view of a solar cell provided according to an embodiment of the present disclosure;

[0027] FIG. 2 is a partial enlarged view of 1 shown in FIG. 1;

[0028] FIG. 3 is another partial enlarged view of 1 shown in FIG. 1;

[0029] FIG. 4 is a schematic view of carrier transport in the solar cell provided according to an embodiment of the present disclosure;

[0030] FIG. 5 is a schematic structural view of a top view of another solar cell provided according to an embodiment of the present disclosure;

[0031] FIG. 6 is a schematic structural view of a top view of yet another solar cell provided according to an embodiment of the present disclosure;

[0032] FIG. 7 is a schematic structural view of a top view of still another solar cell provided according to an embodiment of the present disclosure;

[0033] FIG. 8 is a schematic structural view of a top view of a solar cell provided according to another embodiment of the present disclosure;

[0034] FIG. 9 is a partial enlarged view of 2 shown in FIG. 8;

[0035] FIG. 10 is a schematic structural view of a top view of another solar cell provided according to another embodiment of the present disclosure;

[0036] FIG. 11 is a schematic structural view of a top view of yet another solar cell provided according to another embodiment of the present disclosure;

[0037] FIG. 12 is still another partial enlarged view of 1 shown in FIG. 1;

[0038] FIG. 13 is yet still another partial enlarged view of 1 shown in FIG. 1;

[0039] FIG. 14 is a schematic structural view of a photovoltaic module provided according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0040] It is known from the background technology that the photoelectric conversion efficiency of a solar cells in the prior art is low.

[0041] It is found in the analysis that one of the reasons for the low photoelectric conversion efficiency of the solar cell in the prior art is that, at present, in order to prevent light from being absorbed by a doped conductive layer, the doped conductive layer is generally disposed in a metallized area, and the doped conductive layer in a non-metallized area will be thinned or removed. However, this will lead to the lack of lateral transport channels for carriers in a substrate, that is, the carriers in the substrate are more transported to the area covered by the doped conductive layer, while the area without the doped conductive layer lacks the transport of the carriers, so that the filling factor of the solar cell is greatly reduced, resulting in a problem that the overall power generation efficiency of the solar cell is low.

[0042] A solar cell is provided according to the embodiments of the present disclosure, in which multiple doped conductive layers are disposed at intervals, so that in response to incident light irradiating an area between two adjacent doped conductive layers, since the area has no doped conductive layers, the incident light in the area will not be absorbed by the multiple doped conductive layers, thereby reducing the parasitic absorption of the incident light by the multiple doped conductive layers and improving the utilization rate of the incident light by the substrate. In addition, the conductive transport layer is arranged between two adjacent doped conductive layers and in contact with the doped conductive layer, so that the majority carriers in the substrate can be transported into the multiple doped conductive layers through the conductive transport layer, so that the lateral transport of the majority carriers in the substrate is improved, the filling factor of the solar cell is improved, so as to improve the transport capability of the majority carriers in the substrate while improving the utilization rate of incident light, thereby generally improving the photoelectric conversion efficiency of the solar cell.

[0043] The embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings. However, those skilled in the art may appreciate that, in the various embodiments of the present disclosure, numerous technical details are set forth in order to provide the reader with a better understanding of the present disclosure. However, the technical solutions claimed in the present disclosure may be implemented without these technical details and various changes and modifications based on the following embodiments.

[0044] FIG. 1 is a schematic structural view of a top view of a solar cell provided according to an embodiment of the present disclosure. FIG. 2 is a partial enlarged view of 1 shown in FIG. 1. FIG. 3 is another partial enlarged view of 1 shown in FIG. 1. FIG. 4 is a schematic view of carrier transport in the solar cell provided according to an embodiment of the present disclosure.

[0045] Referring to FIG. 1, FIG. 2 and FIG. 4, a solar cell **110** includes: a substrate **100**; a tunneling dielectric layer **101** disposed over a first surface of the substrate **100**; multiple doped conductive layers **102** arranged at intervals over the tunneling dielectric layer **101**; multiple first electrodes **103** each extending in a first direction X, where the multiple first electrodes **103** respectively correspond to the multiple doped conductive layers **102** and are arranged at intervals along a second direction Y perpendicular to the first direction X, and each of the multiple first electrodes **103** is disposed on and electrically connected to a corresponding one of the multiple doped conductive layers **102**; at least one conductive transport layer **104** located between every two adjacent doped conductive layers **102** in the multiple doped conductive layers **102**, and in contact with a side surface of each of the two adjacent doped conductive layers **102**.

[0046] The conductive transport layer **104** is arranged between every two adjacent doped conductive layers **102** in the multiple doped conductive layers **102** and in contact with the doped conductive layer **102**, so that the majority carriers in the substrate **100** can be transported to the multiple doped conductive layer **102** through the conductive transport layer **104**. In this way, the lateral transport of majority carriers in the substrate **100** is improved, the filling factor of the solar cell **110** is improved, the utilization rate of incident light is improved, and the transport capability

of the majority carriers in the substrate **100** is improved, thereby improving the overall photoelectric conversion efficiency of solar cell **110**. For details, referring to FIG. **4**, which is a schematic view of carrier transport in the solar cell provided according to an embodiment of the present disclosure. With the arrangement of the conductive transport layer **104**, the carriers in the substrate **100** can move laterally into a lateral transport layer, and is transported into the multiple doped conductive layers **102** through the lateral transport layer **104**, thereby increasing the transport capability of carriers in the substrate **100** and increasing the carrier concentration in the multiple doped conductive layers **102**.

[0047] In other embodiments, each of the doped conductive layers **102** includes multiple main body portions **10** disposed at intervals, and the multiple main body portions **10** are electrically connected to the first electrode **103**, that is, the thickness of the doped conductive layer **102** in the metallized area is relatively thicker, so that the multiple main body portions **10** can play a role of reducing metal contact recombination. In response to incident light irradiating the area between the adjacent main body portions **10**, the incident light is not absorbed by the multiple main body portions **10**, so that the absorption and utilization rate of the incident light by the substrate **100** can be improved.

[0048] In addition, each of the doped conductive layers **102** further includes a first connecting portion **11**, the first connecting portion **11** is located between every two adjacent main body portions **10**, which forms a lateral transport channel for the carriers, so that the majority carriers in the substrate **100** can be transported into the main body portion **10** through the first connecting portion **11**, thereby improving the lateral transport capability of the carriers in the substrate **100** in the multiple doped conductive layers **102**. In addition, since the main body portion **10** and the first connecting portion **11** are integrally formed, it is beneficial to reduce the loss of carrier transport at the contact interface between the main body portion **10** and the first connecting portion **11** and further improve the carrier transport efficiency.

[0049] The substrate **100** is configured to receive incident light and generate photogenerated carriers. In some embodiments, the substrate **100** may be embodied as a silicon substrate, and the silicon substrate may be made of at least one of monocrystalline silicon, polycrystalline silicon, amorphous silicon, or microcrystalline silicon. In other embodiments, the substrate **100** may also be made of at least one of silicon carbide, organic material, or multicomponent compound, where the multicomponent compound includes perovskite, gallium arsenide, cadmium telluride, copper indium selenide, and the like.

[0050] In some embodiments, the substrate **100** is doped with N-type or P-type doping ions, where the N-type doping ions may be any one of phosphorus (P) element, bismuth (Bi) element, antimony (Sb) element, arsenic (As) element, or other group V elements, the P-type element may be any one of boron (B) element, aluminum (Al) element, gallium (Ga) element, indium (In) element, or other group III elements. For example, in response to the substrate **100** being P-type substrate **100**, the substrate **100** has P-type doping elements. Or, in response to the substrate **100** being N-type substrate **100**, the substrate **100** has N-type doping elements, such as anyone of phosphorus element, bismuth element, antimony element, arsenic element.

[0051] In some embodiments, the solar cell **110** is a tunnel oxide passivated contact (TOPCON) cell, the substrate **100** further includes a second surface opposite to the first surface, the first surface and the second surface of the substrate **100** are both configured to receive incident light or reflected light. In some embodiments, the first surface may be a backside surface of the substrate **100**, and the second surface may be a frontside surface of the substrate **100**. In other embodiments, the first surface may also be the frontside surface of the substrate **100**, and the second surface may be the backside surface of the substrate **100**.

[0052] In some embodiments, the first surface of the substrate **100** may be embodied as a non-pyramid textured surface, such as a stacked step topography, so that the tunneling dielectric layer **101** on the first surface of the substrate **100** has high density and uniformity, which causes the

tunneling dielectric layer **101** has a desirable passivation effect on the first surface of the substrate **100**. The second surface of the substrate **100** may be embodied as a pyramid textured surface, so that the reflectivity of the second surface of the substrate **100** to incident light is lower, resulting in a higher absorption and utilization rate of light.

[0053] In some embodiments, the tunneling dielectric layer **101** and the doped conductive layer **102** are configured to form a passivation contact structure on surfaces of the substrate **100**, so as to reduce the recombination of carriers in the surfaces of the substrate **100**, thereby increasing the open circuit voltage and improving the photoelectric conversion efficiency of the solar cell **110**. Specifically, the tunneling dielectric layer **101** can reduce the concentration of defect states on the first surface of the substrate **100**, so that the number of recombination centers on the first surface of the substrate **100** is reduced, thereby reducing the recombination rate of carriers.

[0054] The doped conductive layer **102** is configured to form a field passivation layer, so that minority carriers escape from the interface, thereby reducing the concentration of minority carriers. Since the carrier recombination rate at the interface of the substrate **100** is lower, the open circuit voltage of the solar cell **110** is reduced, the short-circuit current and the filling factor are relatively large, which improves the photoelectric conversion performance of the solar cell **110**. In some embodiments, the doped conductive layer **102** and the substrate **100** have doping elements of the same conductivity type.

[0055] In some embodiments, the multiple doped conductive layers **102** extend along the first direction X, and the multiple doped conductive layers **102** are disposed at intervals along the second direction Y, where the second direction Y is perpendicular to the first direction X. In some embodiments, the first electrodes **103** and the doped conductive layers **102** are in a one-to-one correspondence, that is, one first electrode **103** is electrically connected to one doped conductive layer **102**. That is to say, the doped conductive layer **102** is only provided in the area corresponding to the first electrode **103**, so that the parasitic light absorption effect of the area without the first electrode **103** can be reduced, and the utilization rate of light by the substrate **100** can be improved. In some embodiments, the first electrodes **103** may be made of at least one of silver, aluminum, copper, tin, gold, lead, or nickel.

[0056] In other embodiments, the multiple main body portions **10** extend along the first direction X, and the multiple main body portions **10** are disposed at intervals along the second direction Y, where the second direction Y is perpendicular to the first direction X. In some embodiments, the first electrodes **103** and the main body portions **10** are in a one-to-one correspondence, that is, one first electrode **103** is electrically connected to one main body portion **10**. That is to say, the main body portion **10** is only provided in the area corresponding to the first electrode **103**, so that the parasitic light absorption of incident light done by the doped conductive layer **102** is reduced while improving the contact recombination of the first electrode **103**. In some embodiments, the first electrodes **103** may be made of at least one of silver, aluminum, copper, tin, gold, lead, or nickel.

[0057] The tunneling dielectric layer **101** and the multiple doped conductive layers **102** are stacked. Specifically, in some embodiments, the tunneling dielectric layer **101** covers the entire first surface of the substrate **100**, and the multiple doped conductive layers **102** are disposed at intervals on the top surface of the tunneling dielectric layer **101**. In other embodiments, the tunneling dielectric layer **101** is disposed corresponding to the doped conductive layers **102**, that is, the tunneling dielectric layer **101** is disposed between the doped conductive layer **102** and the substrate **100**, and the tunneling dielectric layer **101** is also located between the conductive transport layer **104** and the substrate **100**, so that a part of the tunneling dielectric layer **101** reduces the recombination of carriers on the first surface of the substrate **100**, thereby increasing the concentration of carriers transported to the conductive transport layer **104**.

[0058] In some embodiments, the tunneling dielectric layer **101** may be made of, but is not limited to, aluminum oxide, silicon oxide, silicon nitride, silicon oxynitride, intrinsic amorphous silicon, intrinsic polysilicon, and other dielectric materials with tunneling function. Specifically, the

tunneling dielectric layer **101** may be formed of a silicon oxide layer including silicon oxide (SiO₂). The silicon oxide has desirable passivation properties, and the carriers can easily tunnel through the silicon oxide layer.

[0059] In some embodiments, the conductive transport layer **104** is made of the same material as the doped conductive layer **102**. By making the conductive transport layer **104** of the same material as the doped conductive layer **102**, on the one hand, types of materials in the entire production process can be reduced to facilitate management. on the other hand, the contact between the conductive transport layer **104** and the doped conductive layer **102** is desirable, so that the transport of carriers at an interface between the doped conductive layer **102** and the conductive transport layer **104** is desirable, thereby reducing transport loss. In addition, the transport rates of carriers in the conductive transport layer **104** and the doped conductive layer **102** can be made similar or the same, thereby improving the transport efficiency of carriers from the conductive transport layer **104** to the doped conductive layer **102**. It is worth noting that the same material here means that the conductive transport layer **104** has the same doping ion type and doping ion concentration as those in the doped conductive layer **102**.

[0060] In other embodiments, the main body portion **10** and the first connecting portion **11** are integrally formed, on the one hand, the types of materials in the whole production process can be reduced, so as to facilitate management. On the other hand, the first connecting portion **11** and the main body portion **10** are made to have the same carrier type and carrier concentration, so that the transport of carriers at an interface between the main body portion **10** and the first connecting portion **11**, thereby reducing transport loss. In addition, the transport rate of carriers in the main body portion **10** and the first connecting portion **11** can be made the same, thereby improving the transport efficiency of carriers from the first connecting portion **11** to the main body portion **10**.

[0061] Specifically, in some embodiments, the doped conductive layer **102** is made of at least one of doped amorphous silicon, doped polysilicon or doped microcrystalline silicon. Correspondingly, the conductive transport layer **104** may also be made of one of doped amorphous silicon, doped polysilicon or doped microcrystalline silicon material.

[0062] It can be understood that, in other embodiments, the conductive transport layer **104** may also be made of different material from the doped conductive layer **102**, for example, the conductive transport layer **104** may be made of one of doped amorphous silicon, doped polysilicon or doped microcrystalline silicon, the doped conductive layer **102** may be made of another one of doped amorphous silicon, doped polysilicon, or doped microcrystalline silicon.

[0063] In some embodiments, in response to the conductive transport layer **104** being made of different material from the doped conductive layer **102**, the absorption coefficient of the conductive transport layer **104** to the incident light can be set to be smaller than the absorption coefficient of the incident light of the conductive transport layer **104**, so that the absorption capability of the conductive transport layer **104** for incident light can be reduced while improving the lateral transport capability of carriers, thereby improving the utilization rate of the incident light by the solar cell **110**.

[0064] In some embodiments, since the conductive transport layer **104** is made of the same material as the doped conductive layer **102**, the actual process method for preparing the doped conductive layer **102** and the conductive transport layer **104** is as follows.

[0065] An initial tunneling dielectric layer **101** and an initial doped conductive layer **102** are formed on the first surface of the substrate **100** by a deposition process, where the initial tunneling dielectric layer **101** covers the entire first surface of the substrate **100**, and the initial doped conductive layer **102** covers the entire first surface of the tunneling dielectric layer **101**.

[0066] A patterning process is performed on the top surface of the initial doped conductive layer **102** to define the shape of the doped conductive layer **102** disposed at intervals and the shape of the conductive transport layer **104**.

[0067] The patterned initial doped conductive layer **102** is subjected to an etching process to

remove a part of the initial doped conductive layer **102** to form the doped conductive layers **102** disposed at intervals and the conductive transport layers **104** located between adjacent doped conductive layers **102**.

[0068] In some embodiments, a laser process is used to perform laser etching on the initial doped conductive layer **102**, so that the etching process is relatively simple, and the patterning process for the initial doped conductive layer **102** may be omitted, which is beneficial to simplify the preparation process.

[0069] In some embodiments, in response to the tunneling dielectric layer **101** being arranged to cover the entire first surface of the substrate **100**, and the multiple doped conductive layers **102** being arranged on the top surface of the tunneling dielectric layer **101** at intervals, in the etching process, only the initial doped conductive layer **102** is etched, and the initial tunneling dielectric layer **101** is served as the tunneling dielectric layer **101**.

[0070] In other embodiments, the tunneling dielectric layer **101** is disposed corresponding to the doped conductive layer **102**, that is, the tunneling dielectric layer **101** is disposed between the doped conductive layer **102** and the substrate **100**, and the tunneling dielectric layer **101** is also located between the conductive transport layer **104** and the substrate **100**, the initial tunneling dielectric layer **101** is etched simultaneously during the process of etching the initial doped conductive layer **102** to form the doped conductive layer **102** and the conductive transport layer **104** corresponds to the tunneling dielectric layer **101**.

[0071] In some embodiments, there are multiple conductive transport layers **104**, and the multiple conductive transport layers **104** are disposed along the first direction X at intervals. By disposing multiple conductive transport layers **104** between two adjacent doped conductive layers **102**, the majority carriers in the substrate **100** can be transported into the doped conductive layers **102** through the multiple conductive transport layers **104**, thereby enhancing the lateral transport capability of majority carriers in the substrate **100**. In addition, the multiple conductive transport layers **104** are disposed in a space manner, that is, the multiple conductive transport layers **104** do not cover all areas between two adjacent doped conductive layers **102**, but are disposed on a partial area between two adjacent doped conductive layers **102**. In this way, in response to the conductive transport layer **104** being made of the same material as the doped conductive layer **102**, the overall area of the conductive transport layer **104** will not be excessive, thereby preventing the incident light from being excessively absorbed by the conductive transport layer **104**, resulting in a low utilization rate of the incident light by the substrate **100**.

[0072] In some embodiments, the multiple conductive transport layers **104** are disposed in an array, the array includes multiple columns of conductive transport layers **104** disposed at intervals along the second direction Y, multiple conductive transport layers **104** in each column of the multiple columns of conductive transport layers **104** are disposed at intervals along the first direction X, there is at least one first electrode **103** between two adjacent columns of conductive transport layers **104** along the second direction Y, and the second direction Y is perpendicular to the first direction X. That is, in some embodiments, in response to only one first electrode **103** being arranged between adjacent conductive transport layers **104**, there is at least one conductive transport layer **104** between every two adjacent first electrodes **103**. In other embodiments, there may also be multiple first electrodes **103** between two adjacent columns of conductive transport layers **104**, so that there is at least one conductive transport layer **104** between some of the two adjacent first electrodes **103**, and there is no conductive transport layer **104** between other adjacent first electrodes. For example, along the first direction X, there is at least one conductive transport layer **104** between a No. 1 first electrode **103** and a No. 2 first electrode **103**, and there is no conductive transport layer **104** between the No. 2 first electrode **103** and a No. 3 first electrode **103**. It can be understood that, in response to the conductive transport layer **104** being made of the same material as the doped conductive layer **102**, the greater the number of the conductive transport layer **104**, the stronger the absorbing capability to incident light while enhancing the lateral capability of carriers.

Therefore, connecting relationship between the conductive transport layer **104** and the doped conductive layer **102** can be flexibly set based on the total number of the first electrodes **103** and the demand for the current collecting capability of the first electrodes **103**, so as to prevent the incident light from being excessively absorbed by the conductive transport layer **104** while improving the transport capability of carriers.

[0073] Referring to FIG. 1, in some embodiments, at least one conductive transport layer **104** is disposed between all adjacent first electrodes **103**, which improves the lateral transport capability between adjacent first electrodes **103**, thereby improving the current collecting capability of each first electrode **103**.

[0074] Referring to FIG. 2, in some embodiments, the top surface of the conductive transport layer **104** is lower than or flush with the top surface of the doped conductive layer **102**. In response to the top surface of the conductive transport layer **104** being arranged no higher than the top surface of the doped conductive layer **102**, the top surface of the conductive transport layer **104** is prevented from extending over the top surface of the doped conductive layer **102**, so that the side surface of the conductive transport layer **104** is prevented from absorbing the incident light, thereby reducing the parasitic absorption capability of the conductive transport layer **104** to incident light. It can be understood that in response to the top surface of the conductive transport layer **104** being lower than the top surface of the doped conductive layer **102**, the top surface of the doped conductive layer **102** will play a certain role of sheltering on the incident light incident obliquely to the top surface of the conductive transport layer **104**. Therefore, the transport capability of the conductive transport layer **104** to incident light can be further reduced. In response to the top surface of the conductive transport layer **104** being flush with the top surface of the doped conductive layer **102**, the production process of the solar cell **110** can be simplified, and the conductive transport layer **104** and the doped conductive layer **102** can be formed in the same step by laser ablation.

[0075] In some embodiments, along the direction perpendicular to the surface of the substrate **100**, the height of the conductive transport layer **104** may be 0.5 to 1.2 times the height of the doped conductive layer **102**, and the specific value may be 0.5, 0.6, 0.7, 0.8, 0.9, 1 or 1.2. Within this range, on the one hand, the thickness of the conductive transport layer **104** will not be excessively small, so that the lateral transport capability of the conductive transport layer **104** to carriers will not be too poor. On the other hand, the thickness of the conductive transport layer **104** is not excessive, so as to prevent the incident light from being excessively absorbed due to excessive thickness of the conductive transport layer **104**. Referring to FIG. 3, in other embodiments, the top surface of the first connecting portion **11** is lower than or flush with the top surface of the main body portion **10**. In response to the top surface of the first connecting portion **11** being lower than the top surface of the main body portion **10**, the top surface of the main body portion **10** will block the incident light obliquely incident on the top surface of the first connecting portion **11** to a certain extent, so that the transport capability of the first connecting portion **11** to incident light can be further reduced. In response to the top surface of the first connecting portion **11** being flush with the top surface of the main body portion **10**, the production process of the solar cell **110** can be simplified, and the first connecting portion **11** and the main body portion **10** can be formed in the same step by laser ablation. In some embodiments, in the direction perpendicular to the surface of the substrate **100**, the height of the first connecting portion **11** may be 0.5 to 1.2 times the height of the main body portion **10**, and the specific value may be 0.5, 0.6, 0.7, 0.8, 0.9, 1 or 1.2.

[0076] Referring to FIG. 5 and FIG. 6, FIG. 5 is a schematic structural view of a top view of another solar cell provided according to an embodiment of the present disclosure, and FIG. 6 is a schematic structural view of a top view of yet another solar cell provided according to an embodiment of the present disclosure. In some embodiments, a column of conductive transport layers **104** and an adjacent column of conductive transport layers **104** are arranged in a stagger manner along the first direction X.

[0077] Specifically, in some embodiments, each conductive transport layer **104** in the first column

of conductive transport layers **104** and each conductive transport layer **104** in the second column of conductive transport layers **104** are not aligned in the second direction Y. That is, each conductive transport layer **104** in the first column of conductive transport layers **104** and each conductive transport layer **104** in the second column of conductive transport layers **104** are staggered in the first direction X. Multiple conductive transport layers **104** are arranged in a stagger manner, on the one hand, the number of conductive transport layers **104** is prevented from being excessive, thereby preventing the conductive transport layers **104** from absorbing more incident light. On the other hand, the conductive transport layers **104** can be uniformly distributed on the first surface of the substrate **100**, while the number of the conductive transport layers **104** is relatively small, so that the lateral transport capability of carriers at different positions in the substrate **100** can be enhanced.

[0078] Referring to FIG. 1, in other embodiments, each conductive transport layer **104** in a column of conductive transport layers **104** is in one-to-one correspondence with each conductive transport layer **104** in an adjacent column of conductive transport layers **104**, and corresponding two conductive transport layers **104** are arranged at intervals along the second direction Y. For example, each conductive transport layer **104** in the first column of conductive transport layers **104** and the corresponding conductive transport layer **104** in the second column of conductive transport layers **104** are aligned and arranged in the second direction Y, and each column of conductive transport layers **104** are aligned and arranged, so that the number of conductive transport layers **104** is increased, thereby forming more lateral transport channels to laterally transport carriers in the substrate **100**. In addition, since the conductive transport layers **104** in each column are aligned and arranged, in the actual process of preparing the conductive transport layers **104**, the process of forming the conductive transport layers **104** can be simplified.

[0079] Referring to FIG. 1, FIG. 5 and FIG. 6, in some embodiments, the solar cell **110** further includes multiple second electrodes **106** arranged at intervals, where the multiple second electrodes **106** extend along the second direction Y, and are electrically connected to the multiple first electrodes arranged at intervals along the second direction Y. The multiple second electrodes **106** are arranged at intervals along the first direction X, and the multiple second electrodes **106** are electrically connected to the multiple first electrodes **103**, so as to collect current in the multiple first electrodes **103**, and the current is lead out of the solar cell **110**. It can be understood that the second electrode **106** is not only in electrical contact with the first electrode **103**, but also in electrical contact with a part of the doped conductive layer **102**, so that the carriers in the doped conductive layer **102** can be directly transported to the multiple second electrodes **106** without passing through the multiple first electrodes **103**, thereby improving the capability of the second electrode **106** to collect current.

[0080] In some embodiments, in a column of conductive transport layers **104**, at least one second electrode **106** is disposed between two adjacent conductive transport layers **104**. That is to say, the second electrode **106** is spaced apart from the conductive transport layer **104**. In this way, the second electrode **106** can be position-limited by the conductive transport layer **104**, so that position of the second electrode **106** can be determined without performing additional positioning during the process of preparing the second electrode **106**, which facilitates the printing of the second electrode **106** and simplifies the process procedure.

[0081] Referring to FIG. 5 to FIG. 6, specifically, in some embodiments, in a column of conductive transport layers **104**, two second electrodes **106** are disposed between two adjacent conductive transport layers **104**. That is to say, the conductive transport layers **104** are sparsely disposed, so that the incident light is prevented from being excessively absorbed by the conductive transport layers **104** due to excessive quantity of the conductive transport layers **104**.

[0082] It can be understood that since the conductive transport layers **104** serve as lateral transport channels for carriers, the carrier concentration in the doped conductive layer **102** adjacent to the conductive transport layers **104** is relatively high, so that a part of the first electrode **103**

electrically connected to the doped conductive layer **102** adjacent to the conductive transport layers **104** has a higher carrier concentration. Based on this, in some embodiments, a column of conductive transport layers **104** and an adjacent column of conductive transport layers **104** are disposed in a stagger manner along the first direction X, and two conductive transport layers **104** belong to different columns of the conductive transport layers **104** and disposed in a stagger manner are located on opposite sides of the second electrode **106**, respectively. The conductive transport layers **104** located on two sides of the second electrode **106** are not aligned in the first direction X. In this way, in response to the number of the conductive transport layers **104** being limited, the conductive transport layers **104** are uniformly distributed on two sides of the second electrode **106**. The conductive transport layers **104** are arranged on two sides of the second electrode **106**, that is, the second electrode **106** is electrically connected to the part of the first electrode **103** with higher carrier concentration, so that the collection capability of the second electrode **106** on current in the first electrode **103** can be integrally improved. In addition, due to the small number of conductive transport layers **104**, the incident light is prevented from being excessively absorbed by the conductive transport layers **104**, thereby improving the overall photoelectric conversion performance of the solar cell **110**.

[0083] It can be understood that, in other embodiments, a projection of a part of the second electrode **106** on the first surface of the substrate **100** may also overlap a part of the conductive transport layer **104**. In this way, the second electrode **106** can cover a part of the top surface of the conductive transport layer **104** to partially shield the conductive transport layer **104**, thereby reducing the parasitic light absorption capability of the conductive transport layer **104** to incident light, and further improving the photoelectric conversion efficiency of the solar cell **110**. In some embodiments, the solar cell **110** further includes a second connecting portion **105**, the second electrode **106** is also in direct electrical contact with the second connecting portion **105** being covered, and the second connecting portion **105** is configured to be a lateral transport channel between adjacent second electrodes **106**, so that the carriers in the second connecting portion **105** and the conductive transport layer **104** are also directly transported to the second electrode **106**, which further improves the current collecting capability of the second electrode **106**.

[0084] It can be understood that, in a step of laminating the solar cell **110**, in order to prevent solar cell pieces from being crushed, the second electrode **106** is generally disposed far from edges of the solar cell pieces, that is, edges of the substrate **100** are spaced from the second electrode **106**, which causes the number of second electrodes **106** at the edges of the substrate **100** to be less, so that the second electrode **106** located at the outermost of the edges has a weaker capability to collect carriers at the edges of the substrate **100**. Based on this, in some embodiments, the substrate **100** includes a peripheral area and a central area, the peripheral area is defined as a periphery of the second electrode **106** located at an outermost side, the central area is defined as an area of the substrate **100** apart from the peripheral area, and a distance between every two adjacent conductive transport layers **104** located in the peripheral area in the first direction X is smaller than a distance between every two adjacent conductive transport layers **104** located in the central area in the first direction X. In this way, the density of the conductive transport layer **104** on the first surface of the substrate **100** in the peripheral area is greater than that in the central area, that is, the lateral transport capability of carriers in the substrate **100** corresponding to the peripheral area is stronger, so that the carrier concentration in the first electrode **103** in the peripheral area is relatively higher, so as to compensate the number of carriers collected by the outermost second electrode **106** and improve the current collecting capability of the outermost second electrode **106**.

[0085] In some embodiments, among the conductive transport layers **104** in each column, there are multiple conductive transport layers **104** in the peripheral area, and there is one conductive transport layer **104** or no conductive transport layer **104** between two adjacent second electrodes **106** in the central area. That is to say, the conductive transport layer **104** is sparsely distributed in the central area, thereby reducing the parasitic light absorption capability of the conductive

transport layer **104** to incident light. In the peripheral area, the conductive transport layer **104** is densely distributed, so as to improve the current collecting capability of the outermost second electrode **106**, thereby further integrally improving the photoelectric conversion performance of the solar cell **110**.

[0086] Specifically, referring to FIG. 5, in some embodiments, among the first column of conductive transport layers **104** in the peripheral area, the number of the conductive transport layers **104** on the outermost second electrode **106** side may be 2, and among the second column of conductive transport layers **104** in the peripheral area, the number of the conductive transport layers **104** on the outermost second electrode **106** side may be 1, and the first column of conductive transport layers **104** are disposed in a stagger manner with the second column of conductive transport layers **104**. Only the arrangement of first column and the second column of the conductive transport layers **104** is shown here, and reference may be made to the first column and the second column for the arrangement of the conductive transport layers **104** in the remaining third, fourth, fifth and sixth columns.

[0087] Referring to FIG. 7, in other embodiments, among the first column of conductive transport layers **104** in the peripheral area, the number of the conductive transport layers **104** on the outermost second electrode **106** side may be one, among the second column of conductive transport layers **104** in the peripheral area, the number of the conductive transport layers **104** on the outermost second electrode **106** side may be one, and among the third column of conductive transport layers **104** in the peripheral area, the number of the conductive transport layers **104** on the outermost second electrode **106** side may be one. The adjacent three columns of the conductive transport layers **104** are disposed in a stagger manner along the first direction X. Only the arrangement of the conductive transport layers **104** in the first, second and third columns is shown here, and reference may be made to the first column, the second column and the third column of conductive transport layers **104** for the arrangement of the conductive transport layers **104** in the remaining columns.

[0088] In some embodiments, in each column of the conductive transport layers **104** located in the central area, a distance between every two adjacent conductive transport layers **104** in the first direction X is constant. In each column of the conductive transport layers **104**, the distance between every two adjacent conductive transport layers **104** is constant, which facilitates the laser ablation process during the formation process, that is, it is not necessary to adjust the distance between every two adjacent conductive transport layers **104**, thereby facilitating the production. In addition, the distance between every two adjacent conductive transport layers **104** in the central area are constant, so that the conductive transport layers **104** in the central area are evenly distributed, thereby uniformly improving the carrier collecting capability of the second electrode **106** at different positions.

[0089] In some embodiments, in each column of the conductive transport layers **104** located in the central area, the distance between every two adjacent conductive transport layers **104** ranges from 0.01 mm to 20 mm, for example, the distance may range from 0.01 mm to 0.1 mm, from 0.1 mm to 0.5 mm, from 0.5 mm to 2 mm, from 2 mm to 5 mm, from 5 mm to 10 mm, from 10 mm to 15 mm, or from 15 mm to 20 mm. in each column of the conductive transport layers located in the peripheral area, the distance between every two adjacent conductive transport layers **104** ranges from 0.005 mm to 18 mm, for example, the distance may range from 0.005 mm to 0.01 mm, from 0.01 mm to 0.1 mm, from 0.1 mm to 0.5 mm, from 0.5 mm to 2 mm, from 2 mm to 5 mm, from 5 mm to 10 mm, from 10 mm to 15 mm, or from 15 mm to 18 mm. Within this range, on the one hand, the distance between adjacent conductive transport layers **104** is not too small, so as to prevent the conductive transport layers **104** from absorbing too much incident light due to excessively dense arrangement of the conductive transport layers **104**; on the other hand, within this range, the distance between every two adjacent conductive transport layers **104** is not too small, so that more lateral transport channels are formed, which can greatly improve the lateral

transport capability of carriers in the substrate **100**.

[0090] Referring to FIG. 6, FIG. 6 is a schematic structural view of a top view of yet another solar cell provided according to an embodiment of the present disclosure. In some embodiments, the solar cell **110** further includes: a second connecting portion **105**, the second connecting portion **105** is located between adjacent conductive transport layers **104** arranged at intervals along the first direction X, and is electrically connected to side surfaces of the two adjacent conductive transport layers **104**. It can be understood that the width of the second connecting portion **105** in the second direction Y is smaller than the distance between two adjacent doped conductive layers **102** in the second direction Y, that is, a side surface of the second connecting portion **105** is not in contact with the side surfaces of the two adjacent doped conductive layers **102**. In this way, in response to incident light irradiating the gap between the doped conductive layer **102** and the second connecting portion **105**, the incident light will not be absorbed by the second connecting portion **105** or the doped conductive layer **102**. In some embodiments, the second connecting portion **105** may be made of the same material as the conductive transport layer **104**, so that the second connecting portion **105** may also serve as a lateral transport channel for carriers in the substrate **100**. Specifically, the carriers in the substrate **100** corresponding to the second connecting portion **105** can be transported to the second connecting portion **105**, the carriers in the second connecting portion **105** are then transported to the conductive transport layer **104**, and the carriers are transported to the doped conductive layer **102** through the conductive transport layer **104**. It is not difficult to find that due to the second connecting portion **105**, more carriers in the substrate **100** can be transported to the conductive transport layer **104** and finally reach the doped conductive layer **102**, thereby improving the lateral transport capability of the carriers in the substrate **100**. Therefore, the carrier concentration in the doped conductive layer **102** is bigger, thereby increasing the current collecting capability of the first electrode **103**.

[0091] Referring to FIG. 8 and FIG. 9, in some other embodiments, the doped conductive layer **102** further includes: a bottom connecting portion **20**, and the bottom connecting portion **20** is located between two adjacent main body portions **10** and is connected to side surfaces of two adjacent main body portions **10**. The top surface of the bottom connecting portion **20** is lower than the top surface of the main body portion **10**, and the first connecting portion **11** is located on a part of the top surface of the bottom connecting portion **20**. That is to say, the thickness of the doped conductive layer **102** on the surface of the substrate **100** corresponding to the non-metallized area is thinner than that of the doped conductive layer **102** on the surface of the substrate **100** corresponding to the metallized area, so that the parasitic absorption of incident light done by the doped conductive layer **102** corresponding to the non-metallized area can be reduced. In addition, the bottom connecting portion **20** located in the non-metallized area is further configured to provide a transport channel for majority carriers between adjacent main body portions **10**.

[0092] It can be seen from the above analysis that for the solution in which the doped conductive layer **102** only includes the main body portion **10**, the doped conductive layer **102** corresponding to the non-metallized area is removed, and for the solution in which the doped conductive layer **102** further includes the bottom connecting portion **20**, the doped conductive layer **102** corresponding to the non-metallized area is thinned, so that the carrier transport capability of the substrate **100** corresponding to the non-metallized area is relatively weak. Based on this, the first connecting portion **11** is arranged between the two adjacent main body portions **10** to provide a lateral transport channel between the two adjacent main body portions **10** for majority carriers, so that the transport efficiency of carriers in the substrate **100** and between the doped conductive layers **102** is increased, thereby improving the filling factor of the solar cell **110** and the photoelectric conversion efficiency of the solar cell **110**.

[0093] The tunneling dielectric layer **101** and the main body portions **10** are stacked. Specifically, in some embodiments, the tunneling dielectric layer **101** covers the entire first surface of the substrate **100**, and the multiple main body portions **10** are disposed at intervals on the top surface

of the tunneling dielectric layer **101**. In other embodiments, the tunneling dielectric layer **101** is disposed corresponding to the main body portions **10**, that is, the tunneling dielectric layer **101** is disposed between the main body portions **10** and the substrate **100**, and the tunneling dielectric layer **101** is also located between the conductive transport layer **104** and the substrate **100**, so that a part of the tunneling dielectric layer **101** reduces the recombination of carriers on the first surface of the substrate **100**, thereby increasing the concentration of carriers transported to the conductive transport layer **104**.

[0094] The main body portion **10** and the first connecting portion **11** are integrally formed, on the one hand, the types of materials in the whole production process can be reduced, so as to facilitate management. On the other hand, the first connecting portion **11** and the main body portion **10** are made to have the same carrier type and carrier concentration, so that the transport of carriers at an interface between the main body portion **10** and the first connecting portion **11**, thereby reducing transport loss. In addition, the transport rate of carriers in the main body portion **10** and the first connecting portion **11** can be made the same, thereby improving the transport efficiency of carriers from the first connecting portion **11** to the main body portion **10**.

[0095] Referring to FIG. 5 and FIG. 7, in some embodiments, a column of the first connecting portions **11** and an adjacent column of the first connecting portions **11** are disposed in a stagger manner along the first direction X. Specifically, in some embodiments, each first connecting portions **11** in the first column of first connecting portions **11** and each first connecting portions **11** in the second column of first connecting portions **11** are not aligned in the second direction Y, that is each first connecting portions **11** in the first column of first connecting portions **11** and each first connecting portions **11** in the second column of first connecting portions **11** are arranged in a stagger manner in the first direction X. By arranging multiple first connecting portions **11** in a stagger manner, on the one hand, the number of first connecting portions **11** is prevented from being excessive, thereby preventing the first connecting portions **11** from absorbing too much incident light. On the other hand, the first connecting portions **11** can be uniformly distributed on the first surface of the substrate **100**, while the number of the conductive transport layers **104** is relatively small, so that the lateral transport capability of carriers at different positions in the substrate **100** can be enhanced.

[0096] In some embodiments, each first connecting portion **11** in a column of first connecting portions **11** is in one-to-one correspondence with each first connecting portion **11** in an adjacent column of first connecting portions **11**, and two corresponding first connecting portions **11** are arranged at intervals along the second direction Y. That is, each first connecting portion **11** in the first column of first connecting portions **11** and the corresponding first connecting portion **11** in the second column of first connecting portions **11** are aligned and arranged in the second direction Y, and each column of first connecting portions **11** are aligned and arranged, so that the number of conductive transport layers **104** is increased, thereby forming more lateral transport channels to laterally transport carriers in the substrate **100**. In addition, since the first connecting portions **11** in each column are aligned and arranged, in the actual process of preparing the first connecting portions **11**, the process of forming the first connecting portions **11** can be simplified.

[0097] Referring back to FIG. 1, FIG. 5 and FIG. 6, in some embodiments, the solar cell **110** further includes multiple second electrodes **106** arranged at intervals, where the multiple second electrodes **106** extend along the second direction Y, and are electrically connected to the multiple first electrodes arranged at intervals along the second direction Y. The multiple second electrodes **106** are arranged at intervals along the first direction X, and the multiple second electrodes **106** are electrically connected to the multiple first electrodes **103**, so as to collect current in the multiple first electrodes **103**, and the current is lead out of the solar cell **110**. It can be understood that the second electrode **106** is not only in electrical contact with the first electrode **103**, but also in electrical contact with a part of the doped conductive layer **102**, so that the carriers in the doped conductive layer **102** can be directly transported to the multiple second electrodes **106** without

passing through the multiple first electrodes **103**, thereby improving the capability of the second electrode **106** to collect current.

[0098] In some embodiments, the second electrodes **106** and the first connecting portions **11** are arranged at intervals, or a projection of the first connecting portion **11** on the substrate **100** at least partially overlaps a projection of the second electrode **106** on the substrate **100**. By arranging the second electrodes **106** and the first connecting portions **11** at intervals, the second electrode **106** can be position-limited by the first connecting portion **11**, so that position of the second electrode **106** can be determined without performing additional positioning during the process of preparing the second electrode **106**, which facilitates the printing of the second electrode **106** and simplifies the process flow.

[0099] The projection of the first connecting portion **11** on the substrate **100** at least partially overlaps the projection of the second electrode **106** on the substrate **100**, that is to say, a part of the second electrodes **106** can cover a part of the top surface of the first connecting portion **11** to shield a part of the first connecting portion **11**, so as to reduce the parasitic light absorption capability of the first connecting portion **11** to incident light, thereby further improving the photoelectric conversion efficiency of the solar cell **110**. In some embodiments, the second electrode **106** is also in direct electrical contact with the covered first connecting portion **11**, since the first connecting portion **11** and the main body portion **10** are integrally formed, and the first connecting portion **11** and the main body portion **10** are both in electrical contact with the second electrodes **106**, so that a lateral transport channel is also formed between the adjacent second electrodes **106**, the carriers in the first connecting portion **11** can also be directly transported to the second electrodes **106**, and thereby further improving the current collecting capability of the second electrodes **106**.)

[0100] It can be understood that since the first connecting portions **11** serve as lateral transport channels for carriers, the carrier concentration in the main body portions **10** adjacent to the first connecting portions **11** is relatively high, so that a part of the first electrode **103** electrically connected to the main body portions **10** adjacent to the first connecting portions **11** has a higher carrier concentration. Based on this, in some embodiments, a column of first connecting portions **11** and an adjacent column of first connecting portions **11** are disposed in a stagger manner along the first direction X, and two first connecting portions **11** belong to different columns of the first connecting portions **11** and disposed in a stagger manner are located on opposite sides of the second electrode **106**, respectively. The first connecting portions **11** located on two sides of the second electrode **106** are not aligned in the first direction X. In this way, in response to the number of the first connecting portions **11** being limited, the first connecting portions **11** are uniformly distributed on two sides of the second electrode **106**. The first connecting portions **11** are arranged on two sides of the second electrode **106**, that is, the second electrode **106** is electrically connected to the part of the first electrode **103** with higher carrier concentration, so that the collection capability of the second electrode **106** on current in the first electrode **103** can be integrally improved. In addition, due to the small number of first connecting portions **11**, the incident light is prevented from being excessively absorbed by the first connecting portions **11**, thereby improving the overall photoelectric conversion performance of the solar cell **110**.

[0101] It can be understood that, in a step of laminating the solar cell **110**, in order to prevent solar cell pieces from being crushed, the second electrode **106** is generally disposed far from edges of the solar cell pieces, that is, edges of the substrate **100** are spaced from the second electrode **106**, which causes the number of second electrodes **106** at the edges of the substrate **100** to be less, so that the second electrode **106** located at the outermost of the edges has a weaker capability to collect carriers at the edges of the substrate **100**. Based on this, in some embodiments, the substrate **100** includes a peripheral area and a central area, the peripheral area is defined as a periphery of the second electrode **106** located at an outermost side, the central area is defined as an area of the substrate **100** apart from the peripheral area, and a distance between every two adjacent first connecting portions **11** located in the peripheral area in the first direction X is smaller than a

distance between every two adjacent first connecting portions **11** located in the central area in the first direction X. In this way, the density of the first connecting portions **11** on the first surface of the substrate **100** in the peripheral area is greater than that in the central area, that is, the lateral transport capability of carriers in the substrate **100** corresponding to the peripheral area is stronger, so that the carrier concentration in the first electrode **103** in the peripheral area is relatively higher, so as to compensate the number of carriers collected by the outermost second electrode **106** and improve the current collecting capability of the outermost second electrode **106**.

[0102] In some embodiments, in the first connecting portions **11** of each column, the number of the first connecting portions **11** located in the peripheral area is multiple, and in the central area, the first connection between two adjacent second electrodes **106** The number of the parts **11** is one or zero.

[0103] Specifically, referring to FIG. **8**, in some embodiments, among the first column of first connecting portions **11** in the peripheral area, the number of the first connecting portions **11** on the outermost second electrode **106** side may be 2, and among the second column of first connecting portions **11** in the peripheral area, the number of the first connecting portions **11** on the outermost second electrode **106** side may be 1, and the first column of first connecting portions **11** are disposed in a stagger manner with the second column of first connecting portions **11**. Only the arrangement of first column and the second column of the first connecting portions **11** is shown here, and reference may be made to the first column and the second column for the arrangement of the first connecting portions **11** in the remaining third, fourth, fifth and sixth columns.

[0104] Referring to FIG. **10**, in other embodiments, among the first column of first connecting portions **11** in the peripheral area, the number of the first connecting portions **11** on the outermost second electrode **106** side may be one, among the second column of first connecting portions **11** in the peripheral area, the number of the first connecting portions **11** on the outermost second electrode **106** side may be one, and among the third column of first connecting portions **11** in the peripheral area, the number of the first connecting portions **11** on the outermost second electrode **106** side may be one. The adjacent three columns of the first connecting portions **11** are disposed in a stagger manner along the first direction X. Only the arrangement of the first connecting portions **11** in the first, second and third columns is shown here, and reference may be made to the first column, the second column and the third column of first connecting portions **11** for the arrangement of the first connecting portions **11** in the remaining columns.

[0105] In some embodiments, in each column of the first connecting portions **11** located in the central area, the distance between every two adjacent first connecting portions **11** ranges from 0.01 mm to 20 mm, for example, the distance may range from 0.01 mm to 0.1 mm, from 0.1 mm to 0.5 mm, from 0.5 mm to 2 mm, from 2 mm to 5 mm, from 5 mm to 10 mm, from 10 mm to 15 mm, or from 15 mm to 20 mm. in each column of the first connecting portions **11** located in the peripheral area, the distance between every two adjacent first connecting portions **11** ranges from 0.005 mm to 18 mm, for example, the distance may range from 0.005 mm to 0.01 mm, from 0.01 mm to 0.1 mm, from 0.1 mm to 0.5 mm, from 0.5 mm to 2 mm, from 2 mm to 5 mm, from 5 mm to 10 mm, from 10 mm to 15 mm, or from 15 mm to 18 mm. Within this range, on the one hand, the distance between adjacent first connecting portions **11** is not too small, so as to prevent the first connecting portions **11** from absorbing too much incident light due to excessively dense arrangement of the first connecting portions **11**; on the other hand, within this range, the distance between every two adjacent first connecting portions **11** is not too small, so that more lateral transport channels are formed, which can greatly improve the lateral transport capability of carriers in the substrate **100**.

[0106] Referring to FIG. **12** to FIG. **13**, in some embodiments, the top surface of the conductive transport layer **104** has a light trapping structure **108**. The light trapping structure **108** is configured to enhance the reflection capability of the top surface of the conductive transport layer **104** to the incident light, so that the incident light irradiating on the top surface of the conductive transport layer **104** can be reflected and prevented from being absorbed by the conductive transport layer

104. This part of the reflected incident light can continue to be reflected back, for example, this part of the reflected incident light can be reflected to the area not covered by the doped conductive layer **102** and the conductive transport layer **104**, so as to be absorbed and utilized by the substrate **100**. In this way, the absorption and utilization rate of the substrate **100** to the incident light can be enhanced.

[0107] Specifically, referring to FIG. **12**, in some embodiments, the light trapping structure **108** includes multiple pyramid structures, each of the multiple pyramid structures has a bottom surface and a side surface connected to the bottom surface. Incident light may be repeatedly reflected between side surfaces of two adjacent pyramid structures, so as to reflect the incident light irradiating on the top surface of the conductive transport layer **104** to out of the conductive transport layer **104**, thereby reducing the absorption of the incident light done by the conductive transport layer **104**. Moreover, since each of the pyramid structures has multiple side surfaces, reflection probability of the incident light is further increased, the absorption of the incident light done by the conductive transport layer **104** is further reduced. The reflected incident light can be re-reflected to a part of the first surface of the substrate **100** not covered by the doped conductive layer **102** and the conductive transport layer **104**, which increases the utilization rate of incident light by the substrate **100**, increases the open circuit voltage and short circuit current, thereby improving the photoelectric conversion efficiency of the solar cell **110**.

[0108] In other embodiments, the light trapping structure **108** further includes a recessed structure that is recessed toward the substrate **100**, and the recessed structure is provided, on the one hand, the top surface of the conductive transport layer **104** is lower than the top surface of the doped conductive layer **102**, the doped conductive layer **102** has a certain shielding effect on the incident light irradiating on the top surface of the conductive transport layer **104**. On the other hand, the incident light is enabled to be repeatedly reflected on sidewalls of the recessed structure, thereby reducing the parasitic absorption of the incident light by the top surface of the doped conductive layer **102**.

[0109] Specifically, in some embodiments, along a direction in which the doped conductive layer **102** points to the center of the recessed structure, the height of the recessed structure gradually decreases. Referring specifically to FIG. **12**, the recessed structure has two opposite sidewalls, the tops of the two opposite sidewalls are spaced apart from each other, and the bottoms are connected, that is, the two sidewalls of the recessed structure are inclined relative to the first surface of the substrate **100**. In this way, in response to the incident light irradiating one of two sidewall surfaces, one part of the incident light will be reflected from a first sidewall surface to a second sidewall surface, and after that, in the incident light reflected to the second sidewall surface, one part of the incident light will be reflected to the outside, and the other part of the incident light will be re-reflected from the second sidewall surface to the first sidewall surface. In this way, after the incident light is reflected multiple times, the incident light is substantially emitted to the outside, so that the incident light emitted to the outside has a high probability of being re-reflected to a part of the first surface the substrate **100** that is not covered by the doped conductive layer **102** and the conductive transport layer **104**.

[0110] In other embodiments, referring to FIG. **13**, in the direction in which the doped conductive layer **102** points to the substrate **100**, the cross-sectional shape of the recessed structure may also be a rectangle. That is, the recessed structure has two opposite side walls and a bottom wall, the two opposite side walls are perpendicular to the first surface of the substrate **100**, and the bottom wall can be arranged parallel to the first surface of the substrate **100**.

[0111] It can be understood that, in other embodiments, the recessed structure may also be in other shapes, as long as the recessed structure recesses toward the substrate **100**.

[0112] Referring to FIG. **2**, in some embodiments, the solar cell **110** further includes: a first passivation layer **107**, a part of the first passivation layer **107** covers the first surface of the substrate **100**, and the remaining part of the first passivation layer **107** covers the doped conductive

layer **102** and the top surface of the conductive transport layer **104**. That is to say, the tunneling dielectric layer **101** is disposed correspondingly to the doped conductive layer **102**. The tunneling dielectric layer **101** is disposed between the doped conductive layer **102** and the substrate **100**, and between the conductive transport layer **104** and the substrate **100**, so that the tunneling dielectric layer **101** only covers a part of the surface of the substrate **100**, and a part of the first passivation layer **107** can directly in contact with the first surface of the substrate **100**. Since the conductive transport layer **104** is disposed between every two adjacent doped conductive layers **102**, multiple lateral transport channels are formed in the substrate **100** in direct contact with the first passivation layer **107**, and the carriers in the substrate **100** can move laterally into the doped conductive layer **102**, thereby reducing the consumption of carriers in the transport process and increasing the transport rate. Moreover, since the doped conductive layers **102** are arranged at intervals and are only arranged in the metallized area (the area corresponding to the first electrode **103**), in response to incident light irradiating the area between the adjacent doped conductive layers **102**, the probability of incident light being absorbed is greatly reduced, and the parasitic absorption of incident light by the doped conductive layer **102** is integrally reduced. It can be seen from this that the solar cell **110** provided according to the embodiments of the present disclosure not only improve the utilization rate of the incident light by the solar cell **110**, but also maintain a relatively high transport efficiency of carriers in the solar cell **110**.

[0113] In some embodiments, the first passivation layer **107** may be a single-layer or multi-layer structure, and the first passivation layer **107** may be made of at least one of magnesium fluoride, silicon oxide, aluminum oxide, silicon oxynitride, silicon nitride, titanium oxide.

[0114] In other embodiments, a front surface of the tunneling dielectric layer **101** may also be disposed on the first surface of the substrate **100**. Based on this, the first passivation layer **107** is disposed to partially cover the top surface of the doped conductive layer **102** and the top surface of the conductive transport layer **104**. The remaining part of the first passivation layer **107** covers the first surface of the substrate **100**.

[0115] In some embodiments, the first passivation layer **107** may be formed by using a plasma enhanced chemical vapor deposition (PECVD) method after the doped conductive layer **102** and the conductive transport layer **104** are formed.

[0116] The first electrode **103** penetrates through the first passivation layer **107** to be electrically connected to the doped conductive layer **102**. The first passivation layer **107** is configured to reduce the reflection of the incident light by the substrate **100**. In some embodiments, after the first passivation layer **107** is formed, multiple first electrodes **103** disposed at intervals are formed on a side of the doped conductive layer **102** away from the substrate **100**, the first electrode **103** extends along the first direction X, and is electrically connected to the doped conductive layer **102**.

[0117] In some embodiments, the second surface of the substrate **100** has an emitter with a type of doping ions in the emitter that is different from the type of doping ions in the doped conductive layer **102**. In some embodiments, a surface of the emitter away from the substrate **100** further has an anti-reflection layer, and the anti-reflection layer is configured to anti-reflect incident light. In some embodiments, the anti-reflection layer may be a silicon nitride layer, and the silicon nitride layer is made of silicon nitride material. In other embodiments, the anti-reflection layer may also be provided in a multi-layer structure, for example, may be a stacked-layer structure composed of one or more materials selected from silicon nitride, silicon oxide, or silicon oxynitride.

[0118] In other embodiments, the second surface of the substrate **100** also has a structure similar to structures formed on the first surface of the substrate **100**, for example, the second surface of the substrate **100** may have a second tunneling dielectric layer and a second doped conductive layer stacked in sequence along a direction away from the second surface of the substrate **100**. The type of doping ions in the second doped conductive layer is different from the type of doping ions in the doped conductive layer **102**.

[0119] In some embodiments, the solar cell **110** further includes a third electrode (not shown). The

third electrode is located on the second surface of the substrate **100**. In response to the second surface of the substrate **100** having an emitter, the third electrode penetrates through the anti-reflection layer to be electrically connected to the emitter. In response to the second surface of the substrate **100** being formed with structures similar to that of the first surface of the substrate **100**, the third electrode is electrically connected to the second doped conductive layer.

[0120] In the solar cell **110** provided according to the above embodiments, the conductive transport layer **104** is arranged between every two adjacent doped conductive layers **102** in the multiple doped conductive layers **102** and in contact with the doped conductive layer **102**, so that the majority carriers in the substrate **100** can be transported to the multiple doped conductive layer **102** through the conductive transport layer **104**. In this way, the lateral transport of majority carriers in the substrate **100** is improved, the filling factor of the solar cell **110** is improved, the utilization rate of incident light is improved, and the transport capability of the majority carriers in the substrate **100** is improved, thereby improving the overall photoelectric conversion efficiency of solar cell.

[0121] Correspondingly, a photovoltaic module is further provided according to an embodiment of the present disclosure. Referring to FIG. **14**, the photovoltaic module includes at least one cell string, where the at least one cell string is formed by connecting multiple solar cells **110**, each of the multiple solar cells **110** being a solar cell **110** according to any one above; at least one package layer **120** configured to cover a surface of the at least one cell string; at least one cover plate **130** configured to cover a surface of the at least one package layer **120** away from the at least one cell string. The solar cells **110** are electrically connected in the form of a whole piece or multiple pieces to form multiple cell strings, and the multiple cell strings are electrically connected in series and/or parallel.

[0122] Specifically, in some embodiments, the multiple cell strings are electrically connected by conductive strips **140**. The package layer **120** covers the front surface and the back surface of the solar cell **110**. Specifically, the package layer **120** may be embodied as an ethylene-vinyl acetate copolymer (EVA) adhesive film, a polyethylene octene co-elastomer (POE) adhesive film, a polyethylene terephthalate (PET) film, or other organic package films. In some embodiments, the cover plate **130** may be embodied as a cover plate **130** with a light-transmitting function, such as a glass cover plate, a plastic cover plate, or the like. Specifically, a surface of the cover plate **130** facing the package layer **120** may be a surface with protrusions and recess, thereby increasing the utilization rate of incident light.

[0123] Although the present disclosure is disclosed above with preferred embodiments, it is not used to limit the claims. Any person skilled in the art can make some possible changes and modifications without departing from the concept of the present disclosure. The scope of protection shall be subject to the scope defined by the claims of the present disclosure.

[0124] Those of ordinary skill in the art can understand that the above embodiments are specific examples for realizing the present disclosure, and in actual disclosures, various changes may be made in form and details without departing from the spirit and range of the present disclosure. Any person skilled in the art can make their own changes and modifications without departing from the spirit and scope of the present disclosure. Therefore, the protection scope of the present disclosure should be subject to the scope defined by the claims.

Claims

1. A solar cell, comprising: a substrate; a tunneling dielectric layer disposed over a first surface of the substrate; a plurality of doped conductive layers arranged at intervals over the tunneling dielectric layer; a plurality of first electrodes each extending in a first direction, wherein the plurality of first electrodes are arranged at intervals along a second direction perpendicular to the first direction, and each first electrode of the plurality of first electrodes is disposed on and electrically connected to a corresponding one of the plurality of doped conductive layers; and at

least one conductive transport layer, wherein the at least one conductive transport layer includes a respective conductive transport layer between every two adjacent doped conductive layers in the plurality of doped conductive layers, and the respective conductive transport layer is in contact with a side surface of each of the two adjacent doped conductive layers; wherein the at least one conductive transport layer and the plurality of doped conductive layers are doped with doping ions of a same type.

2. The solar cell according to claim 1, wherein the respective conductive transport layer is disposed between a corresponding pair of adjacent first electrodes of the plurality of first electrodes, and is spaced apart from each of the corresponding pair of adjacent first electrodes.

3. The solar cell according to claim 1, wherein each first electrode is partially embedded in the corresponding one of the plurality of doped conductive layers.

4. The solar cell according to claim 1, wherein there are a plurality of conductive transport layers disposed at intervals along the first direction.

5. The solar cell according to claim 1, wherein the at least one conductive transport layer is configured as a plurality of conductive transport layers, the plurality of conductive transport layers are disposed in a form of an array, the array comprises a plurality of columns of conductive transport layers disposed at intervals along the second direction, conductive transport layers in each column of the plurality of columns of conductive transport layers are disposed at intervals along the first direction, and there is at least one first electrode between two adjacent columns of conductive transport layers along the second direction.

6. The solar cell according to claim 5, wherein conductive transport layers in adjacent columns of conductive transport layers are arranged in a staggered manner along the first direction.

7. The solar cell according to claim 5, wherein each of a column of conductive transport layers is in one-to-one correspondence to each of an adjacent column of conductive transport layers, and every two adjacent conductive transport layers corresponding to each other are disposed in the spaced manner along the second direction.

8. The solar cell according to claim 5, further comprising: a plurality of second electrodes disposed at intervals, wherein each of the plurality of second electrodes extends along the second direction and is electrically connected to the plurality of first electrodes disposed at intervals along the second direction.

9. The solar cell according to claim 8, wherein in a column of the conductive transport layers, there is at least one second electrode between two adjacent conductive transport layers.

10. The solar cell according to claim 9, wherein in the column of the conductive transport layers, there are two second electrodes between two adjacent conductive transport layers.

11. The solar cell according to claim 10, wherein conductive transport layers in adjacent columns of conductive transport layers are arranged in a staggered manner along the first direction, and two conductive transport layers belong to different columns of conductive transport layers and disposed in a stagger manner are located on opposite sides of the second electrode, respectively.

12. The solar cell according to claim 8, wherein the substrate comprises a peripheral area and a central area, the peripheral area is defined as a periphery of the second electrode located at an outermost side, the central area is defined as an area of the substrate apart from the peripheral area, and a distance between every two adjacent conductive transport layers located in the peripheral area in the first direction is smaller than a distance between every two adjacent conductive transport layers located in the central area in the first direction.

13. The solar cell according to claim 12, wherein among conductive transport layers in each column of conductive transport layers, there are multiple conductive transport layers in the peripheral area, and there is at most one conductive transport layer between two adjacent second electrodes in the central area.

14. The solar cell according to claim 12, wherein in each column of the conductive transport layers located in the central area, a distance between every two adjacent conductive transport layers in the

first direction is constant.

15. The solar cell according to claim 14, wherein, in each column of the conductive transport layers located in the central area, the distance between every two adjacent conductive transport layers ranges from 0.01 mm to 20 mm, in each column of the conductive transport layers located in the peripheral area, a distance between every two adjacent conductive transport layers ranges from 0.005 mm to 18 mm.

16. The solar cell according to claim 1, further comprising: a first connecting portion, wherein the first connecting portion is located between two adjacent conductive transport layers disposed at intervals along the first direction, and is electrically connected to side surfaces of the two adjacent conductive transport layers.

17. The solar cell according to claim 1, wherein in the direction perpendicular to the surface of the substrate, a height of a respective one of the at least one conductive transport layer is 0.5 to 1.2 times a height of a corresponding one of the plurality of doped conductive layers.

18. The solar cell according to claim 1, wherein a top surface of the at least one conductive transport layer has a light trapping structure, and the light trapping structure includes multiple pyramid structures, each of the multiple pyramid structures has a bottom surface and a side surface connected to the bottom surface.

19. The solar cell according to claim 1, further comprising: a first passivation layer, one part of the first passivation layer covers the first surface of the substrate, and the other part of the first passivation layer covers top surfaces of the plurality of doped conductive layers and a top surface of the at least one conductive transport layer.

20. A photovoltaic module, comprising: at least one cell string, wherein each of the at least one cell string is formed by connecting a plurality of solar cells, each of the plurality of solar cells being a solar cell according to claim 1; at least one package layer configured to cover a surface of the at least one cell string; and at least one cover plate configured to cover a surface of the at least one package layer away from the at least one cell string.
